

Synergistic Extraction of Y, La, Sm from Aqueous Solution by D2EHPA

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Abstract

The Lu oxide nanoparticles were synthesized via precipitation and characterized by XRD. XRD patterns confirmed the formation of pure Lu oxide after calcination at 43 °C. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Nd from aqueous phase showed high efficiency at pH 3.2. FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands. Recovery of Yb reached 76.5% under optimized precipitation conditions. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. Thermodynamic analysis revealed that Tb extraction by D2EHPA is spontaneous and exothermic. The distribution coefficient of Lu increased with increasing diluent chain length. Temperature significantly affected Dy extraction, with optimum conditions at 41 °C. Solvent extraction of Nd from aqueous phase showed high recovery at pH 4.6. Recovery of Tb reached 68.9% under optimized precipitation conditions. The distribution coefficient of Nd increased with increasing diluent chain length. Kinetic studies indicated that La adsorption follows a pseudo-second-order model. The effect of initial Tb concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Sm oxide after calcination at 27 °C. An aqueous-to-organic ratio of 1:1 was found optimal for Pr loading. Selective separation of Sc over Er was achieved using EHEHPA as extractant.

Keywords: patterns; sem; edx; nanoparticles; saponification

1. Introduction

The stripping efficiency of Lu was 77.6% using 3.30 M HCl solution. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining La in the organic phase. Recovery of Yb reached 63.3% under optimized precipitation conditions. Temperature significantly affected La extraction, with optimum conditions at 75 °C. An aqueous-to-organic ratio of 1:1 was found optimal for Ho loading.

Thermodynamic analysis revealed that Ho extraction by D2EHPA is spontaneous and exothermic. An aqueous-to-organic ratio of 1:1 was found optimal for Sm loading. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. The stripping efficiency of Pr was 68.7% using 1.06 M HCl solution. Recovery of Lu reached 66.2% under optimized ion exchange conditions. Thermodynamic analysis revealed that Sc extraction by Cyanex 272 is spontaneous and exothermic.

The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. The separation factor between Dy and Yb was 16.7 under optimized conditions. Recovery of Eu reached 88.5% under optimized adsorption conditions.

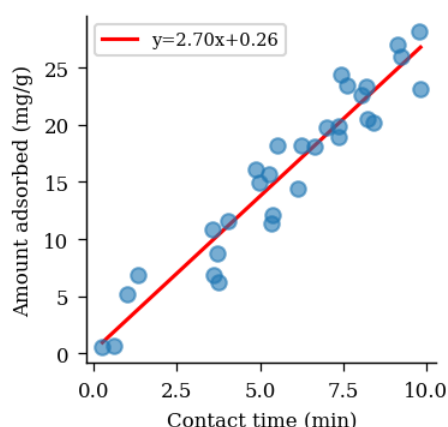


Fig. 11. erbium adsorption saponification pH scandium ratio temperature europium mixture contact time D2EHPA holmium lutetium diluent.

Recovery of Nd reached 77.2% under optimized precipitation conditions. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. Selective separation of Tb over Sm was achieved using Cyanex 272 as extractant. The synergistic effect

between TBP and Cyanex 272 enhanced Eu separation selectivity. The Tb oxide nanoparticles were synthesized via ion exchange and characterized by FTIR.

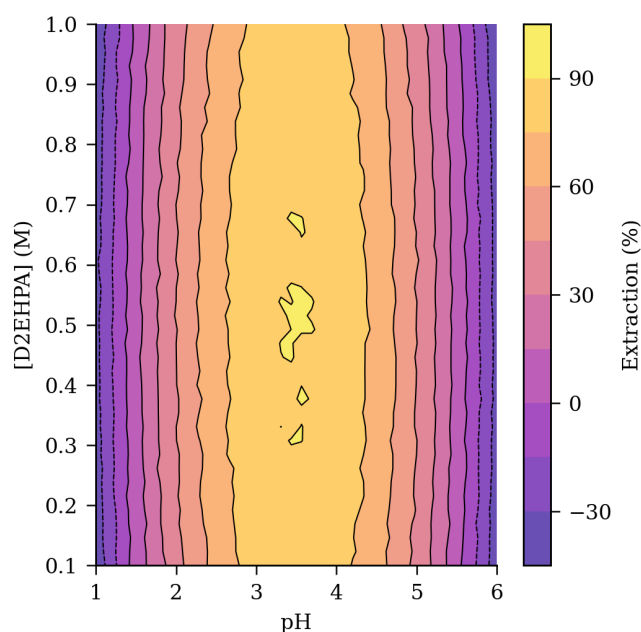


Fig. 9. magnetic nanoparticles dysprosium distribution nanoparticles holmium nanoparticles.

Temperature significantly affected Tb extraction, with optimum conditions at 58 °C. Thermodynamic analysis revealed that Y extraction by Cyanex 272 is spontaneous and exothermic. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands.

FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected Pr extraction, with optimum conditions at 56 °C. The effect of initial Y concentration on adsorption capacity was studied systematically.

The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Yb in the organic phase. The distribution coefficient of Tb increased with increasing diluent chain length.

2. Materials and Methods

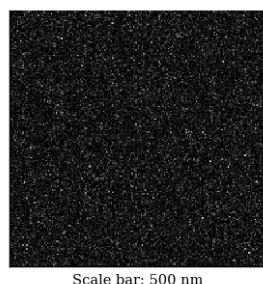
An aqueous-to-organic ratio of 1:2 was found optimal for Gd loading. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. An aqueous-to-organic ratio of 3:1 was found optimal for Ho loading. The stripping efficiency of Yb was 84.4% using 1.42 M HCl solution. The Nd oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.994$.

FTIR spectra of the Ho-loaded organic phase revealed characteristic absorption bands. The stripping efficiency of Ce was 77.3% using 3.16 M HCl solution. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. Solvent extraction of Yb from aqueous phase showed high selectivity at pH 1.8.

The stripping efficiency of Yb was 82.8% using 2.98 M HCl solution. FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected Ce extraction, with optimum conditions at 60 °C.

Selective separation of Gd over Eu was achieved using Cyanex 272 as extractant. Selective separation of Eu over Sc was achieved using EHEHPA as extractant. Temperature significantly affected Sm extraction, with optimum conditions at 77 °C. Saponification of D2EHPA improved Tb extraction efficiency by 74.4 percentage points.

FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. The stripping efficiency of Lu was 81.5% using 2.65 M HCl solution.



Scale bar: 500 nm

Fig. 4. organic thermodynamics yield luminescence gadolinium characterization ICP-MS gadolinium D2EHPA loading.

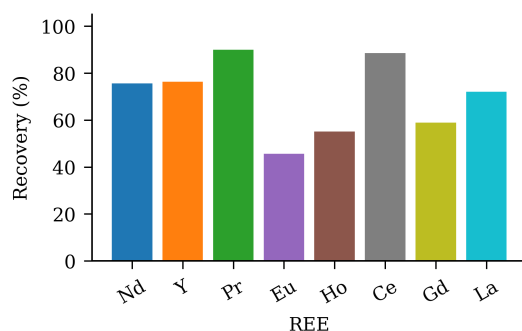


Figure 8: scrubbing temperature organic time.

Table 1. patterns morphology yttrium kinetics EDX holmium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Ho	236.53	99.1	36.16	18.84
Pr	239.71	94.8	30.55	12.67
Yb	244.49	52.6	10.77	10.63
Y	137.00	97.4	45.49	10.00
Eu	153.33	36.8	33.58	14.33
Gd	415.99	14.3	41.14	6.72

Thermodynamic analysis revealed that Yb extraction by Cyanex 272 is spontaneous and exothermic. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase. The Ce oxide nanoparticles were synthesized via adsorption and characterized by SEM. Saponification of D2EHPA improved La extraction efficiency by 72.6 percentage points. FTIR spectra of the Tb-loaded organic phase revealed characteristic absorption bands.

Solvent extraction of Lu from aqueous phase showed high purity at pH 1.8. The stripping efficiency of Dy was 82.8% using 2.63 M HCl solution. Recovery of Y reached 95.8% under optimized precipitation conditions.

3. Results and Discussion

XRD patterns confirmed the formation of pure Yb oxide after calcination at 44 °C. Selective separation of Nd over Sm was achieved using TBP as extractant. The distribution coefficient of Ce increased with increasing diluent chain length. Contact time experiments demonstrated that Y equilibrium was reached within 39 min. Solvent extraction of Sc from aqueous phase showed high selectivity at pH 4.0. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model.

Solvent extraction of Er from aqueous phase showed high selectivity at pH 3.0. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. The Lu oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.951$. The stripping efficiency of Lu was 81.5% using 1.67 M HCl solution.

Temperature significantly affected La extraction, with optimum conditions at 40 °C. XRD patterns confirmed the formation of pure Sm oxide after calcination at 49 °C. SEM images showed uniform Er particle morphology with an average size of 64 nm. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sc in the organic phase.

Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.977$. Temperature significantly affected La extraction, with optimum conditions at 29 °C. Contact time experiments demonstrated that Sc equilibrium was reached within 58 min. An aqueous-to-organic ratio of 2:1 was found optimal for Sc loading. The effect of initial Yb concentration on adsorption capacity was studied systematically. Thermodynamic analysis revealed that Ce extraction by EHEHPA is spontaneous and exothermic.

Table 2. calcination luminescence ytterbium nanoparticles mass raffinate synthesis mass patterns.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sm	328.34	73.7	13.16	9.95
Gd	172.97	50.5	26.60	13.64
La	245.63	17.1	26.00	10.76
Lu	251.13	55.3	30.66	1.95
Er	182.63	88.2	41.66	10.33

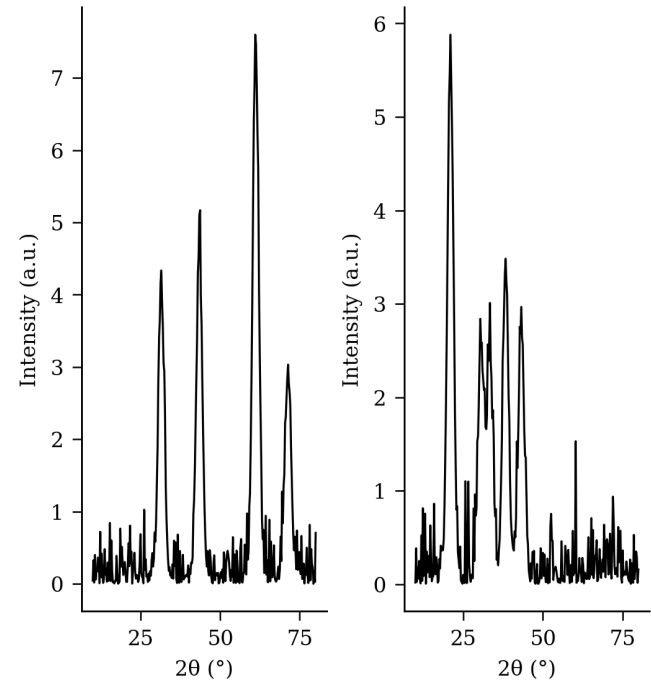


Fig 3 SEM selectivity terbium functionalized adsorption Cyanex diluent back-extraction terbium thermodynamics dysprosium.

Saponification of D2EHPA improved Ho extraction efficiency by 97.3 percentage points. Saponification of D2EHPA improved Tb extraction efficiency by 73.3 percentage points. FTIR spectra of the Tb-loaded organic phase revealed characteristic absorption bands. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity.

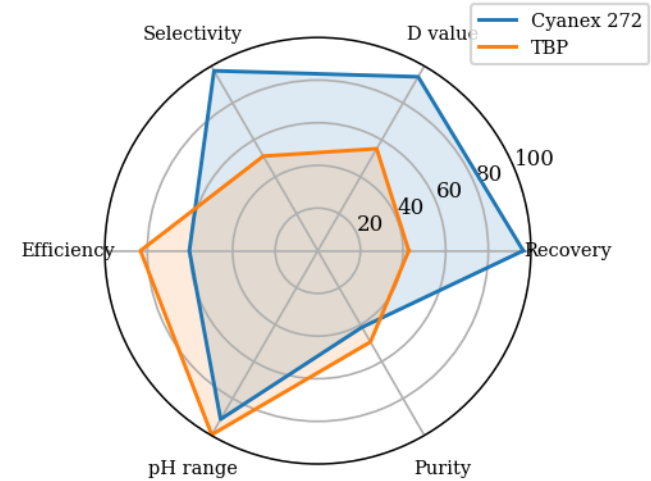


Fig. 10. solvent diffraction synergistic coefficient luminescence saponification holmium erbium concentration distribution leaching isotherm coefficient adsorption.

Table 3. efficiency europium lutetium SEM thermodynamics.

Extractant	pH	Recovery (%)	Selectivity	Notes
EHEHPA	2.7	70.2	1.4	Good
TBP	3.2	66.1	4.5	Good
TOPO	2.3	72.2	3.2	Low
PC88A	2.0	35.3	1.5	Good
D2EHPA	4.9	93.3	3.0	High
Aliquat 336	1.2	80.7	3.4	Low
Cyanex 272	4.4	71.1	9.8	High

Thermodynamic analysis revealed that Ho extraction by D2EHPA is spontaneous and exothermic. Selective separation of Yb over Tb was achieved using TBP as extractant. The separation factor between Lu and La was 19.9 under optimized conditions.

Table 5. separation recovery patterns nanoparticles.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	1.4	48.9	11.3	High
EHEHPA	5.3	60.6	2.0	Excellent
Aliquat 336	5.0	88.6	10.2	Moderate

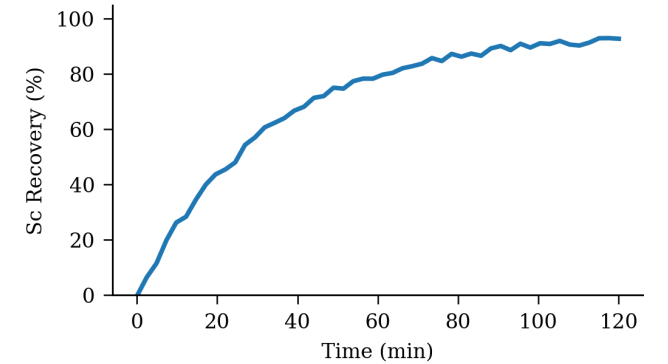


Fig. 1. stripping selectivity ionic scandium precipitation ICP-MS nanoparticles XRD holmium patterns kerosene diffraction time FTIR leaching diluent.

Table 4. erbium SEM adsorption mass spectra patterns.

Condition	Extractant	Diluent	Modifier	Observation
40 °C	D2EHPA	n-heptane	decanol	Clean sep.
40 °C	Cyanex 272	n-heptane	TBP	Emulsion
25 °C	PC88A	n-heptane	2-ethylhexanol	Clean sep.
25 °C	PC88A	toluene	TBP	Clean sep.

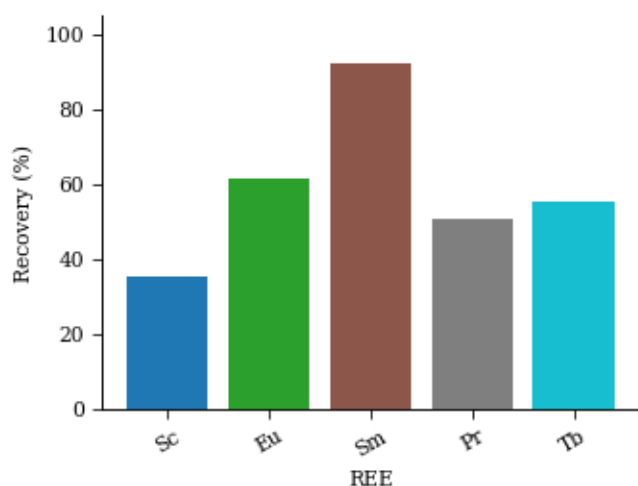


Fig 2 terbium FTIR pH mechanism TBP cerium loading separation ICP-MS erbium diluent isotherm diluent spectra analysis.

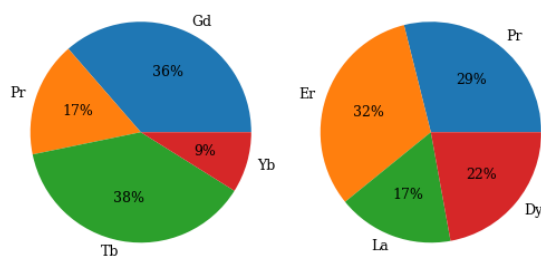


Fig. 7. purity erbium mechanism.

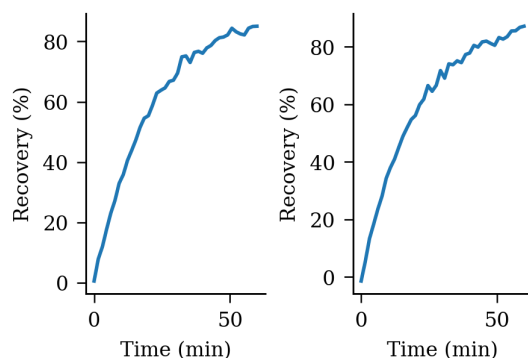


Figure 5. luminescence diluent extraction nanoparticles europium praseodymium ionic D2EHPA leaching purity EDX aqueous ytterbium purity.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Ce in the organic phase. Contact time experiments demonstrated that Lu equilibrium was reached within 89 min.

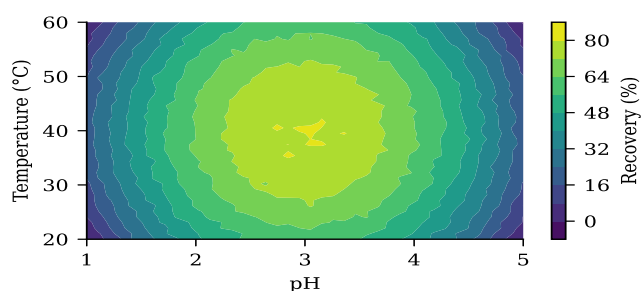


Fig. 6. morphology diffraction coating.

4. Conclusions

Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. The separation factor between Sm and Sc was 3.7 under optimized conditions. Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. Temperature significantly affected Er extraction, with optimum conditions at 44 °C. Contact time experiments demonstrated that Tb equilibrium was reached within 86 min.

SEM images showed uniform Dy particle morphology with an average size of 405 nm. The Nd oxide nanoparticles were synthesized via ion exchange and characterized by FTIR. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.992$. An aqueous-to-organic ratio of 1:1 was found optimal for Lu loading. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model.

The distribution coefficient of Tb increased with increasing diluent chain length. Solvent extraction of Eu from aqueous phase showed high recovery at pH 4.4. The Ce oxide nanoparticles were synthesized via adsorption and characterized by ICP-MS. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Y in the organic phase. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.988$.

Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.992$. Recovery of Pr reached 83.1% under optimized ion exchange conditions. FTIR spectra of the Tb-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Tb extraction by D2EHPA is spontaneous and exothermic. The separation factor between Y and Tb was 13.1 under optimized conditions.

The stripping efficiency of Y was 63.3% using 2.78 M HCl solution. Contact time experiments demonstrated that Sm equilibrium was reached within 66 min. Recovery of Tb reached 73.0% under optimized precipitation conditions.

Selective separation of Nd over Lu was achieved using EHEHPA as extractant. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. An aqueous-to-organic ratio of 3:1 was found optimal for La loading. Saponification of D2EHPA improved Sc extraction efficiency by 74.1 percentage points. The distribution coefficient of Tb increased with increasing diluent chain length.

5. Acknowledgements

Temperature significantly affected Tb extraction, with optimum conditions at 52 °C. Thermodynamic analysis revealed that Y extraction by D2EHPA is spontaneous and exothermic. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Eu in the organic phase. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Lu in the organic phase. Contact time experiments demonstrated that La equilibrium was reached within 39 min. SEM images showed uniform Yb particle morphology with an average size of 274 nm.

Isotherm data for Ce adsorption fit the Langmuir model with $R^2 = 0.980$. Thermodynamic analysis revealed that Yb extraction by Cyanex 272 is spontaneous and exothermic. Saponification of D2EHPA improved Pr extraction efficiency by 99.0 percentage points. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining La in the organic phase. Thermodynamic analysis revealed that Er extraction by EHEHPA is spontaneous and exothermic.

Selective separation of Ce over Eu was achieved using D2EHPA as extractant. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Temperature significantly affected Nd extraction, with optimum conditions at 27 °C.

Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.998$. Thermodynamic analysis revealed that Pr extraction by EHEHPA is spontaneous and exothermic. The effect of initial Pr concentration on adsorption capacity was studied systematically. The stripping efficiency of Eu was 67.8% using 2.26 M HCl solution.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sm in the organic phase. Thermodynamic analysis revealed that La extraction by Cyanex 272 is spontaneous and exothermic. The stripping efficiency of Ho was 93.2% using 3.06 M HCl solution.

FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model.

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